

Mathematics for Data Science

Lecture 2 - Matrices, linear maps and their properties

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Previously covered topics

- Vector space, example of $\mathbb{R}^n$, geometric interpretation
- Subspace (examples and proposition about the dimensions)
- Linear combination, span, linearly independent vectors, spanning list
- Basis, dimension, canonical basis of $\mathbb{R}^n$
- Surjectivity, injectivity, bijectivity, case of linear transformations
- Linear transformations, rank, image/range, kernel/nullspace, rank-nullity theorem
- ML illustrations: representing data as vectors, weight matrices as linear maps etc.

Preliminary remark

Today, we consider matrices with real-valued coefficients.

Notation :

$$\mathcal{M}_{m,n}(\mathbb{R}) = \{A = (a_{ij})_{1 \leq i \leq m, 1 \leq j \leq n} \mid a_{ij} \in \mathbb{R}\}$$

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- 2 Matrices as linear maps
- 3 Range, rank and kernel of a matrix
- 4 A few special matrices
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- 7 Linear systems and Gaussian elimination

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Matrices

To represent and manipulate vectors and linear maps on a computer, we use rectangular arrays of numbers known as **matrices**.

Definition

A matrix is a rectangular array of numbers, called **coefficients**.

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

where A has m rows and n columns. We say A is an $m \times n$ matrix.

Vocabulary : If $m = n$ then A is called a *square matrix*.

Matrix operations

Reminder: $(\mathcal{M}_{m,n}(\mathbb{R}), +, \cdot)$ is a $\mathbb{R}$ -vector space.

Addition of matrices: Let $A = (a_{ij})$ and $B = (b_{ij})$ be two $m \times n$ matrices. Their sum is:

$$A + B = (a_{ij} + b_{ij})_{1 \leq i \leq m, 1 \leq j \leq n}.$$

Multiplication by a scalar: If $\lambda \in \mathbb{R}$ and $A = (a_{ij})$ is an $m \times n$ matrix, then:

$$\lambda A = (\lambda a_{ij})_{1 \leq i \leq m, 1 \leq j \leq n}.$$

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Warning

Matrix addition is only defined when the two matrices have the same size.

Matrix-Vector multiplication

Let $A \in \mathbb{R}^{m \times n}$ and $x \in \mathbb{R}^n$. Then:

$$Ax = \begin{pmatrix} \sum_{j=1}^n a_{1j}x_j \\ \sum_{j=1}^n a_{2j}x_j \\ \vdots \\ \sum_{j=1}^n a_{mj}x_j \end{pmatrix}$$

Interpretation ?

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Interpretation ?

Remark

- *This can be interpreted as a linear combination of the columns of A with weights given by the coordinates of x .*

Matrix-Matrix Multiplication

Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$. Then:

Matrix-Matrix Multiplication

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$$AB = (c_{ij}) \in \mathbb{R}^{m \times p}, \quad c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}.$$

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Be careful about sizes

- Matrix multiplication is only defined if the number of columns of A equals the number of rows of B .

Note: you can come back to matrix-vector multiplication by thinking of matrix B as stacked **column vectors**

Matrix-Matrix multiplication

Proposition

- ① Associativity: Let $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times p}$, $C \in \mathbb{R}^{p \times q}$. Then

$$(AB)C = A(BC).$$

- ② Distributivity (left): If $A \in \mathbb{R}^{m \times n}$ and $B, C \in \mathbb{R}^{n \times p}$, then
 $A(B + C) = AB + AC$.

- ③ Distributivity (right): If $A, B \in \mathbb{R}^{m \times n}$ and $C \in \mathbb{R}^{n \times p}$, then
 $(A + B)C = AC + BC$.

- ④ For any $A \in \mathbb{R}^{m \times n}$, $I_m A = A$ and $A I_n = A$.

Be careful about sizes

- Matrix multiplication is *not* commutative in general, i.e. $AB \neq BA$ in most cases.

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Matrix of a linear map

Proposition

If $A \in \mathbb{R}^{m \times n}$, the following mapping is a linear transformation.

$$\mathbb{R}^n \rightarrow \mathbb{R}^m, \quad x \mapsto Ax$$

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Matrix of a linear map

Suppose V and W are finite-dimensional vector spaces with bases $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ and $\mathcal{C} = (\mathbf{w}_1, \dots, \mathbf{w}_m)$, respectively, and $L : V \rightarrow W$ is a linear map. Then the matrix $A = (a_{ij}) \in \mathbb{R}^{m \times n}$ of L in the bases $\mathcal{B}$ and $\mathcal{C}$ is defined by

$$L(\mathbf{v}_j) = a_{1j} \mathbf{w}_1 + \cdots + a_{mj} \mathbf{w}_m = \sum_{i=1}^m a_{ij} \mathbf{w}_i, \quad j = 1, \dots, n.$$

Matrix of a linear map

NB: the j -th column of $\mathbf{A}$ consists of the coordinates of $L(\mathbf{v}_j)$ in the chosen basis for W .

$$[L(x)]_C = [L]_{C \leftarrow B} [x]_B.$$

Changing the basis in V or W changes the matrix representation.

Notation: next, we also use the following notation

$$[L(x)]_C = [L]_{B,C} [x]_B$$

Matrix of a linear map

Matrix of a linear map (Another formulation)

Let $L : V \rightarrow W$ be a linear transformation between finite-dimensional vector spaces V and W ($\dim(V) = n$, $\dim(W) = m$). If $\{v_1, \dots, v_n\}$ is a basis of V and $\{w_1, \dots, w_m\}$ is a basis of W , then for each $1 \leq j \leq n$, **there exist unique scalars a_{ij} such that**

$$L(v_j) = \sum_{i=1}^m a_{ij} w_i$$

The matrix $A = (a_{ij}) \in \mathbb{R}^{m \times n}$ is the matrix representation of L from the basis $\{v_1, \dots, v_n\}$ to the basis $\{w_1, \dots, w_m\}$.

Remarks

In other words, every matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ induces a linear map $L : \mathbb{R}^n \rightarrow \mathbb{R}^m$ given by (operator notation)

$$L(\mathbf{x}) = \mathbf{A}\mathbf{x}$$

when standard bases are used. More generally, for arbitrary bases,

$$[L(x)]_{\mathcal{C}} = [L]_{\mathcal{C} \leftarrow \mathcal{B}}[x]_{\mathcal{B}}.$$

and the matrix of this map with respect to the standard bases of $\mathbb{R}^n$ and $\mathbb{R}^m$ is $\mathbf{A}$.

Conversely, every linear transformation $L : V \rightarrow W$ with $\dim(V) = n$ and $\dim(W) = m$, can be described by a matrix $A \in \mathbb{R}^{m \times n}$.

Warning

Watch out for the dimensions: $A \in \mathbb{R}^{m \times n}$.

ML illustration: Dense layers are linear maps

In a neural network, a dense layer applies

$$x \in \mathbb{R}^d \longmapsto Wx + b \in \mathbb{R}^k, \quad W \in \mathbb{R}^{k \times d}, \quad b \in \mathbb{R}^k.$$

Linear vs affine

The map $x \mapsto Wx$ is linear. The map $x \mapsto Wx + b$ is affine.

- The matrix W stores the learned linear transformation.
- The bias b translates the output.
- Without nonlinearities, a composition of dense layers is still one affine map.

See for example the implementation of a linear layer in PyTorch¹.

¹<https://pytorch.org/docs/stable/generated/torch.nn.Linear.html>

ML illustration: Processing examples in batches

Let's assume an input $x \in \mathbb{R}^{d_{in}}$ goes through a linear layer parametrized by a weight matrix $W \in \mathbb{R}^{d \times d_{in}}$. Basically, we compute the output as $y = Wx$.

Now, let's assume we have a batch of B input samples $x_1, x_2, \dots, x_B$. We can efficiently implement the operation as a matrix-matrix product by stacking the examples in a matrix of size $d_{in} \times B$.

$$Y = WX, \quad X = [x_1 | x_2 | \dots | x_B] \in \mathbb{R}^{d_{in} \times B}$$

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Reminder: Range, Rank, Kernel of a linear map

Let $L : V \rightarrow W$ be a linear map, V and W vector spaces.

Range of a linear map

The range of L is the set of vectors $y \in W$ such that there exist a vector $x \in V$ that is mapped to y by L .

$$L(V) = \{y \in W \mid y = L(x), x \in V\}$$

Rank of a linear map

The **rank** of L is defined as $\text{rank}(L) = \dim(L(V))$.

Kernel, Nullspace

We define the **kernel** of L as $\text{null}(L) = \{x \in V \mid L(x) = \mathbf{0}_W\}$ (also denoted $\ker(L)$).

Range of a linear map

With matrices, we now have a new way of writing the range (image) of a linear map:

Definition

Let $L : V \rightarrow W$ be a linear map represented by a matrix A . Two equivalent definitions of the range of L are:

$$\text{range}(L) = \{w \in W \mid \exists v \in V, L(v) = w\}$$

$$\text{range}(L) = \{w \in W \mid \exists v \in V, Av = w\}$$

Note: operator notation $/$ or replace with the matrix notation

Range and Rank of a matrix

- The **columnspace** (resp. **rowspace**) of a matrix $A \in \mathbb{R}^{m \times n}$ is the span of its columns, considered as vectors of $\mathbb{R}^m$ (resp. rows, considered as vectors of $\mathbb{R}^n$).
- The column space of A is also the range of the linear map

$$x \in \mathbb{R}^n \mapsto Ax \in \mathbb{R}^m.$$

- The rowspace of A is the range of the linear map from $\mathbb{R}^m$ to $\mathbb{R}^n$ which is induced by $A^\top$.

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Proposition

The dimension of the columnspace of A is the same as the dimension of the rowspace of A and it is called the **rank** of A .

$$\text{rank}(A) = \dim(\text{range}(A)) = \dim(\text{range}(A^\top)).$$

Kernel (Null space) and range (image)

We also have a new way of writing the kernel of a linear map.

Definition

Let $L : V \rightarrow W$ be a linear map represented by a matrix A . We define the kernel/nullspace of L as:

$$\ker(L) = \text{null}(L) = \{v \in V \mid L(v) = 0_W\}.$$

$$\ker(A) = \{x \in V \mid Ax = 0_W\}.$$

Link to the properties of injectivity/surjectivity

Kernel and injectivity

Let $A \in \mathbb{R}^{m \times n}$. The mapping $x \mapsto Ax$ is injective if and only if $\ker(A) = \{0\}$.

Equivalent statements for a square matrix

If $A \in \mathbb{R}^{n \times n}$, the following are equivalent:

- 1 The transformation $x \mapsto Ax$ is bijective.
- 2 $\text{Im}(A) = \mathbb{R}^n$ (i.e. A is surjective).
- 3 $\ker(A) = \{0\}$ (i.e. A is injective).
- 4 $\text{rank}(A) = n$

Rank-nullity theorem for matrices

Theorem (Rank-nullity theorem for matrices)

Let $A \in \mathbb{R}^{m \times n}$. We have the following equality:

$$n = \text{rank}(A) + \dim(\ker(A))$$

NB: Here n is the dimension of the (input) domain $\mathbb{R}^n$, i.e. the number of columns of A .

ML illustration: (reminder) Range and kernel of a linear layer

Let $W \in \mathbb{R}^{k \times d}$ be the weight matrix of a linear layer:

$$x \mapsto Wx.$$

Range: the set of outputs that the linear layer can produce.

$$\text{range}(W) = \{Wx : x \in \mathbb{R}^d\} \subseteq \mathbb{R}^k.$$

Kernel: contains the input directions that are ignored by the layer.

$$\ker(W) = \{x \in \mathbb{R}^d : Wx = 0\}.$$

ML illustration: Kernel as lost information

Consider

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Then

$$A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 + x_2 \\ x_1 + x_2 \end{pmatrix}.$$

- The matrix keeps the sum $x_1 + x_2$.
- It loses the difference $x_1 - x_2$.

We can compute:

$$\ker(A) = \text{span} \left\{ \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}.$$

NB: two inputs that differ by a vector in $\ker(A)$ have the same output.

ML illustration: LoRA and low-rank updates

Large neural networks contain very large weight matrices. Instead of fine-tuning all entries of a matrix W_0 , LoRA² uses

$$W_{\text{new}} = W_0 + \Delta W, \quad \Delta W = BA,$$

where

$$B \in \mathbb{R}^{d \times r}, \quad A \in \mathbb{R}^{r \times k}, \quad r \ll \min(d, k).$$

²<https://arxiv.org/abs/2106.09685>

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Therefore

$$\text{rank}(\Delta W) \leq r.$$

Interpretation

LoRA restricts the trainable update to a low-dimensional subspace.

²<https://arxiv.org/abs/2106.09685>

ML illustration: LoRA parameter count

Suppose we apply LoRA with $r = 8$ to a weight matrix $W_0 \in \mathbb{R}^{4096 \times 4096}$.
Compute the number of parameters to update in the case of a full (dense) update vs the low-rank update.

³https://huggingface.co/docs/peft/main/en/conceptual_guides/lora

ML illustration: LoRA parameter count

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Compute the number of parameters to update in the case of a full (dense) update vs the low-rank update.

A full update ΔW has

$$4096^2 = 16\,777\,216$$

trainable parameters.

With LoRA rank $r = 8$:

$$\Delta W = BA,$$

where

$$B \in \mathbb{R}^{4096 \times 8}, \quad A \in \mathbb{R}^{8 \times 4096}.$$

The number of trainable parameters is

$$4096 \cdot 8 + 8 \cdot 4096 = 65\,536.$$

This is about 256 times fewer parameters. ³

³https://huggingface.co/docs/peft/main/en/conceptual_guides/lora

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Diagonal matrices

Definition

A **diagonal matrix** in $\mathbb{R}^{n \times n}$ is a square matrix of the form

$$D = \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{bmatrix},$$

where $d_1, \dots, d_n \in \mathbb{R}$.

In other words, all coefficients outside the main diagonal are zero.

Note: it is also denoted for short, $\text{diag}(d_1, d_2, \dots, d_n)$

Properties: $D = D^\top$.

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In other words, all coefficients outside the main diagonal are zero.

Note: it is also denoted for short, $\text{diag}(d_1, d_2, \dots, d_n)$

Properties: $D = D^\top$. We will also see that a diagonal matrix $D = \text{diag}(d_1, \dots, d_n)$ is invertible if and only if $d_i \neq 0$ for all i (see lecture on determinant).

Identity matrix

Definition

The **identity matrix** of size n , denoted I_n , is the diagonal matrix of size $n \times n$ with all diagonal entries equal to 1:

$$I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}.$$

Properties of the Identity Matrix

Let A be an $n \times n$ matrix. Then:

- ① $I_n \mathbf{x} = \mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$.
- ② $A I_n = I_n A = A$
- ③ In particular, $I_n I_n = I_n$ and $(I_n)^{-1} = I_n$

ML illustration: Diagonal matrices as feature rescaling

A diagonal matrix acts coordinate by coordinate:

$$D = \text{diag}(d_1, \dots, d_n), \quad Dx = \begin{pmatrix} d_1 x_1 \\ \vdots \\ d_n x_n \end{pmatrix}.$$

Interpretation: Diagonal matrices model independent rescaling of features or channels.

Examples:

- feature normalization;
- learned scale parameters in normalization layers;

Learn more : batch normalization⁴

⁴<https://arxiv.org/abs/1502.03167>

Transpose of a matrix

Definition

If $\mathbf{A} \in \mathbb{R}^{m \times n}$, its *transpose* $\mathbf{A}^\top \in \mathbb{R}^{n \times m}$ is given by $(\mathbf{A}^\top)_{ij} = \mathbf{A}_{ji}$ for each (i, j) .

In other words, the columns of $\mathbf{A}$ become the rows of $\mathbf{A}^\top$, and the rows of $\mathbf{A}$ become the columns of $\mathbf{A}^\top$.

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Properties of the Transpose

Let $\mathbf{A}, \mathbf{B}$ be matrices of compatible sizes and $\alpha \in \mathbb{R}$:

- i) $(\mathbf{A}^\top)^\top = \mathbf{A}$
- ii) $(\mathbf{A} + \mathbf{B})^\top = \mathbf{A}^\top + \mathbf{B}^\top$
- iii) $(\alpha \mathbf{A})^\top = \alpha \mathbf{A}^\top$
- iv) $(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top$

Symmetric matrices

Definition

A matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is said to be **symmetric** if it is equal to its own transpose ($\mathbf{A} = \mathbf{A}^\top$), meaning that $A_{ij} = A_{ji}$ for all (i, j) .

Note: skew-symmetric matrix $\mathbf{A} = -\mathbf{A}^\top$

Note: In Lecture 4 we will remind the spectral theorem, e.g. for real-valued square matrices “If $\mathbf{A} \in \mathbb{R}^{n \times n}$ is symmetric, then there exists an orthonormal basis of $\mathbb{R}^n$ consisting of eigenvectors of $\mathbf{A}$.”

The practical application of this theorem is a particular factorization of symmetric matrices, referred to as the **eigendecomposition** or **spectral decomposition**.

ML illustration: Symmetric matrices in ML

Symmetric matrices naturally appear in ML.

- If the entries in the column vector x are random variables $(X_1, X_2, \dots, X_n)$, each with finite variance and expected value, then the *covariance matrix* K_{xx} is the matrix whose (i,j) entry is the covariance

$$K_{xx}(i,j) = \text{cov}[X_i, X_j] = E[(X_i - E[X_i])(X_j - E[X_j])]$$

with E the expected value.

- Similarity matrices:

$$S_{ij} = \text{sim}(x_i, x_j).$$

NB: Why symmetry matters? Real symmetric matrices admit an orthonormal eigenbasis (see next lectures).

ML illustration: Sparse matrices

A **sparse** matrix contains many zero coefficients. Some examples in ML:

- bag-of-words and TF-IDF matrices in NLP
- attention masks (read more about low-rank structure⁵)
- recommender-system user/item matrices (because a given user rates only a small number of items).
- adjacency matrices of graphs⁶

For a graph with adjacency matrix $A(*)$,

$$A_{ij} = 1 \iff \text{there is an edge between nodes } i \text{ and } j.$$

⁵https://openaccess.thecvf.com/content/WACV2025/papers/Viti_A_0-Shot_Self-Attention_Mechanism_for_Accelerated_Diagonal_Attention_WACV_2025_paper.pdf

⁶Graph neural networks use this matrix to propagate information between neighboring nodes. <https://arxiv.org/abs/1609.02907>

ML illustration: Toeplitz matrices and convolution

A one-dimensional convolution can be written as matrix multiplication.
For example, with kernel (a, b, c) :

$$\begin{pmatrix} a & b & c & 0 & 0 \\ 0 & a & b & c & 0 \\ 0 & 0 & a & b & c \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix}.$$

The matrix has repeated diagonals: it is **Toeplitz**-like.

interpretation: Convolutional layers are not arbitrary dense matrices. They use strong structure: local connectivity and shared coefficients. (more in the DL course⁷)

⁷<https://cs231n.github.io/convolutional-networks/> 

Other particular matrices

- matrix of ones: useful for summation
- Triangular matrix
- Band matrix
- Block-diagonal matrix
- Shift matrix, circulant matrix
- Orthogonal matrices: more in the next lectures
- ...

More examples [here](#).

Exercise

Let $A, B \in \mathbb{R}^{n \times n}$ be real-valued symmetric matrices. Suppose that AB is symmetric.

- 1 Show that A and B commute.
- 2 Deduce that any two diagonal matrices commute.

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Invertible matrix

Definition

Let A be an $n \times n$ matrix. If there exists a matrix $B \in \mathbb{R}^{n \times n}$ such that

$$AB = BA = I_n,$$

then A is said to be **invertible** (or **nonsingular**), and B is called the **inverse** of A . The inverse of A is denoted by A^{-1} .

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Caution

Not every square matrix is invertible.

Invertible matrix

Alternative definition through the lens of linear maps

Definition

Let A be an $n \times n$ matrix. If the linear transformation $x \mapsto Ax$ is bijective, we say that A is *invertible* and denote its inverse by A^{-1} . It satisfies:

$$A^{-1}A = AA^{-1} = I_n$$

where I_n is the $n \times n$ identity matrix.

Properties of invertible matrices

Proposition

If A and B are invertible $n \times n$ matrices, and $\alpha \in \mathbb{R} \setminus \{0\}$, then:

- ① $(A^{-1})^{-1} = A$
- ② $(AB)^{-1} = B^{-1}A^{-1}$
- ③ $(A^{\top})^{-1} = (A^{-1})^{\top}$
- ④ $(\alpha A)^{-1} = \frac{1}{\alpha}A^{-1}$
- ⑤ $I_n^{-1} = I_n$

Theorem of Equivalent Statements for an Invertible Matrix

Theorem

Let $A \in \mathbb{R}^{n \times n}$. The following statements are equivalent:

- 1 *Invertibility: A is invertible.*
- 2 *Trivial kernel: $\ker(A) = \{\mathbf{0}\}$.*
- 3 *Full rank: $\text{rank}(A) = n$.*
- 4 *The columns (resp. rows) of A are linearly independent.*
- 5 *Span: The columns of A span $\mathbb{R}^n$, i.e. $\text{range}(A) = \mathbb{R}^n$.*
- 6 *Linear map: A is bijective as a linear transformation $\mathbb{R}^n \rightarrow \mathbb{R}^n$.*
- 7 *A is surjective as a linear transformation $\mathbb{R}^n \rightarrow \mathbb{R}^n$.*
- 8 *Lin. stm.: For any $\mathbf{b} \in \mathbb{R}^n$, equation $A\mathbf{x} = \mathbf{b}$ has a unique solution.*
- 9 *Equivalence to identity: A is row-equivalent^a to the identity matrix I_n .*
- 10 *Determinant: $\det(A) \neq 0$ (see next lecture on determinant).*

^aTwo matrices are said to be **row equivalent** if one can be changed to the other by a sequence of elementary row operations.

ML illustration: Invertibility and information loss

A square matrix A is invertible when the map

$$x \mapsto Ax$$

loses no information.

Example:

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/3 \end{pmatrix}.$$

The map stretches space, and the inverse undoes the stretching.

By contrast,

$$B = \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix}$$

has rank 1, so it collapses the plane onto a line and is not invertible.

NB: Some generative models are built from invertible transformations⁸

⁸Read more on the principle of normalizing flows

<https://arxiv.org/abs/1807.03039>

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Theorem of basis change

Theorem

Let $\mathcal{B} = (u_1, u_2, \dots, u_n)$ and $\mathcal{B}' = (v_1, v_2, \dots, v_n)$ be two bases of a vector space E , $L : E \rightarrow E$ a linear map, $A = [L]_{\mathcal{B}}$ the matrix of L in $\mathcal{B}$ and $B = [L]_{\mathcal{B}'}$ the matrix of L in $\mathcal{B}'$.

Let P be the matrix such that the j^{th} column is $[v_j]_{\mathcal{B}}$, the coordinates of basis vector v_j of $\mathcal{B}'$ in the basis $\mathcal{B}$.

$$P = [[v_1]_{\mathcal{B}} \dots [v_n]_{\mathcal{B}}], \text{ so that } [x]_{\mathcal{B}} = P[x]_{\mathcal{B}'}$$

Then P is invertible and we have

$$B = P^{-1}AP.$$

Theorem of basis change

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Then P is invertible and we have

$$B = P^{-1}AP.$$

Definition (Similar matrices (FR: *matrices semblables*.))

With the above notations, A and B are called **similar** matrices.

Interpretation: they represent the same endomorphism from $L : V \rightarrow V$ under different choices of basis for V . (only defined for square matrices!)

Theorem of basis change

Same statement, other notations:

- ① $[L]_{\mathcal{B}'} = P^{-1}[L]_{\mathcal{B}}P$
- ② We can also write that $P_{\mathcal{B}'}^{\mathcal{B}}$ is the matrix to change from $\mathcal{B}$ to $\mathcal{B}'$, i.e. $[x]_{\mathcal{B}'} = P_{\mathcal{B}'}^{\mathcal{B}}[x]_{\mathcal{B}}$ and $[x]_{\mathcal{B}} = P_{\mathcal{B}}^{\mathcal{B}'}[x]_{\mathcal{B}'}$.

Then

$$P_{\mathcal{B}}^{\mathcal{B}'} P_{\mathcal{B}'}^{\mathcal{B}} = I_n.$$

and

$$[L]_{\mathcal{B}'} = P_{\mathcal{B}'}^{\mathcal{B}} [L]_{\mathcal{B}} P_{\mathcal{B}}^{\mathcal{B}'}$$

- ③ Alternatively

$$\mathcal{M}_{\mathcal{B}'}^{\mathcal{B}}(\text{Id}_E) \mathcal{M}_{\mathcal{B}}^{\mathcal{B}'}(\text{Id}_E) = \mathcal{M}_{\mathcal{B}'}^{\mathcal{B}'}(\text{Id}_E) = I_n.$$

$$\mathcal{M}_{\mathcal{B}'}^{\mathcal{B}'}(L) = P_{\mathcal{B}'}^{\mathcal{B}} \mathcal{M}_{\mathcal{B}}^{\mathcal{B}}(L) P_{\mathcal{B}}^{\mathcal{B}'}$$

$$\mathcal{M}_{\mathcal{B}'}^{\mathcal{B}'}(L) = \mathcal{M}_{\mathcal{B}'}^{\mathcal{B}}(\text{Id}_E) \mathcal{M}_{\mathcal{B}}^{\mathcal{B}}(L) \mathcal{M}_{\mathcal{B}}^{\mathcal{B}'}(\text{Id}_E)$$

ML illustration: Similarity as change of basis

Changing coordinates can make a problem easier without changing the underlying transformation.

Example of PCA: a covariance matrix is expressed in a basis of eigenvectors, where it becomes diagonal⁹.

$$\Sigma = U\Lambda U^{\top}.$$

The columns of U are principal directions, and Λ stores the variances along those directions.

⁹<https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.PCA.html>

Similar matrices VS equivalent matrices

Definition (Equivalent matrices)

Two matrices $A, B \in \mathbb{R}^{m \times n}$ are called **equivalent** if there exist invertible matrices $Q \in \mathbb{R}^{m \times m}$ and $P \in \mathbb{R}^{n \times n}$ such that

$$B = QAP.$$

Interpretation: Equivalent matrices represent the same linear map after changing the basis in the domain and the basis in the codomain.

Equivalent vs Similar matrices (FR: *mat. semblables vs équivalentes*)

By contrast, similar matrices are square matrices of the form

$$B = P^{-1}AP.$$

i.e., similar matrices represent the same endomorphism after changing one basis in the same vector space.

Similar matrices are equivalent (taking $Q = P^{-1}$), but not conversely.

Orthogonal matrix¹⁰

Definition

A square matrix U is called *orthogonal* if:

$$U^{\top} U = U U^{\top} = I.$$

Equivalently, $U^{-1} = U^{\top}$.

Interpretation: the columns (and rows) of an orthogonal matrix $U \in \mathbb{R}^{n \times n}$ form an orthonormal basis of $\mathbb{R}^n$

Determinant of an orthogonal matrix

(see lecture on determinant) The determinant of a real-valued orthogonal matrix is either 1 or -1 .

¹⁰more in the next lectures

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Linear systems

Definition

The general form of a **system of linear equations** in the unknowns $x_1, \dots, x_n$ is:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases}$$

where $a_{ij} \in \mathbb{R}$ and $b_i \in \mathbb{R}$ are called coefficients.

Voc: Every n -tuple $(x_1, \dots, x_n) \in \mathbb{R}^n$ that satisfies all equations is called a *solution* of the system.

Example of linear systems

- A system of equations **without a solution**.
- A system with a **unique solution**.
- A system with **redundancy** (infinitely many solutions).

Geometric interpretation in 2D

In dimension 2, each equation corresponds to a line in the plane.

- If the lines intersect at a point $\Rightarrow$ unique solution.
- If the lines are parallel and disjoint $\Rightarrow$ no solution.
- If the lines coincide $\Rightarrow$ infinitely many solutions.

Matrix Formulation

The system can be written as a matrix-vector product:

$$Ax = b$$

where

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}.$$

ML illustration: Solving linear systems in regression

Consider a linear model

$$f(t) = at + b.$$

If we want the line to pass through $(1, 2)$ and $(3, 6)$, then

$$a + b = 2, \quad 3a + b = 6.$$

In matrix form:

$$\begin{pmatrix} 1 & 1 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 2 \\ 6 \end{pmatrix}.$$

Solving gives

$$a = 2, \quad b = 0.$$

This is supervised learning in its simplest exact form.

ML illustration: Least squares as an overdetermined system

Usually, we have more equations than unknowns:

$$X\beta \approx y, \quad X \in \mathbb{R}^{n \times d}, \quad n > d.$$

There may be no exact solution to

$$X\beta = y.$$

Instead, linear regression solves

$$\hat{\beta} = \arg \min_{\beta \in \mathbb{R}^d} \|X\beta - y\|_2^2.$$

When $X^\top X$ is invertible:

$$\hat{\beta} = (X^\top X)^{-1} X^\top y.$$

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¹¹https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LinearRegression.html

Solving linear systems

Example

Particular solution of system (also called **special solution**)

Method for finding a general solution :

- 1 Find a particular solution to $Ax = b$.
- 2 Find all solutions to $Ax = 0$.
- 3 Combine the solutions from steps 1. and 2. to the general solution.

Note: A particular solution is generally not unique. The solution set is unique, but it can be parametrized in different ways.

Elementary transformations of linear systems

Elementary transformations : keep the solution set the same, but transform the equation system into a simpler form.

- Exchange of two equations
- Addition of one equation to another
- Multiplication of an equation by a scalar $\lambda \in \mathbb{R}^*$

Elementary transformations of linear systems

Elementary transformations : keep the solution set the same, but transform the equation system into a simpler form.

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- Multiplication of an equation by a scalar $\lambda \in \mathbb{R}^*$

Gaussian elimination is a constructive algorithmic way for transforming any system of linear equations into a particular, simpler form called the **row-echelon form**.

Gaussian elimination

Definition (Row-echelon form)

A matrix is in row-echelon form if

- 1 All rows that contain only zeros are at the bottom of the matrix; correspondingly, all rows that contain at least one nonzero element are on top of rows that contain only zeros.
- 2 Looking at nonzero rows only, the **pivot** (also called **leading coefficient**) of each row is strictly to the right of the pivot of the previous row.
- 3 All entries below a pivot are zero.

NB: We apply elimination to the augmented matrix $[A \mid I_n]$. If it can be reduced to $[I_n \mid B]$, then $B = A^{-1}$. If the left block cannot be reduced to I_n , then A is not invertible.

The row-echelon form makes it easier to determine a particular solution.

Gaussian Elimination

Idea: Reduce the system $Ax = b$ to an equivalent triangular system using elementary row operations. See the [MML book](#) for a detailed exercise. In particular, practice with the examples of Section 2.1.

Gaussian elimination

Definition (Reduced Row Echelon Form)

The matrix describing an equation system is in **reduced row-echelon form** (also: row-reduced echelon form or row canonical form) if

- 1 It is in row-echelon form.
- 2 Every pivot is 1.
- 3 The pivot is the only nonzero entry in its column.

Uniqueness of the reduced row-echelon form

Every $m \times n$ matrix is row-equivalent to a unique reduced row-echelon form.

Calculating an Inverse Matrix by Gaussian Elimination

How to calculate an inverse Matrix by Gaussian elimination ?

Calculating an Inverse Matrix by Gaussian Elimination

How to calculate an inverse Matrix by Gaussian elimination ?

We apply elimination to the augmented matrix $[A \mid I_n]$ to obtain $[I_n \mid A^{-1}]$. If we bring the augmented equation system into reduced row-echelon form, we can read out the inverse on the right-hand side of the equation system. Hence, determining the inverse of a matrix is equivalent to solving systems of linear equations.

Connection with the rank

Rank of a matrix A : The maximum number of linearly independent rows (or columns) of A .

Interpretation in Gaussian Elimination:

- The number of *pivots* obtained in row echelon form equals $\text{rank}(A)$.
- The rank determines whether a system has zero, one, or infinitely many solutions.

Cases:

- If $\text{rank}(A) < \text{rank}([A|b])$: the system is **inconsistent** (no solution).
- If $\text{rank}(A) = \text{rank}([A|b]) = n$: the system has a **unique solution**.
- If $\text{rank}(A) = \text{rank}([A|b]) < n$: the system has **infinitely many solutions**.

Properties of the reduced row echelon form

Pivot columns of A (in the *original* A) form a basis of the column space.
The nonzero rows of the reduced row-echelon form form a basis of the row space.

Note: The number of free variables in a linear system is also called **nullity**, cf rank nullity theorem. For a system of n equations with n unknowns, with matrix A , $n = \text{number of free variables} + \text{rank}(A)$.

Next class

Recap and practice :

- For next week: **Practice with the extra exercise sheets** <https://github.com/EvaJF/math4DS/blob/main/test/quiz0.pdf>
- Optional exercises (good for getting started) <https://prismia.chat/shared/linear-algebra> (except Dot products and orthogonality)
- Further reading (if you need to practice solving linear systems!): <https://mml-book.github.io/book/mml-book.pdf>
- The Matrix Cookbook <https://www.math.uwaterloo.ca/~hwolkowi/matrixcookbook.pdf>
- Examples with executable code: https://d2l.ai/chapter_preliminaries/linear-algebra.html

Next lecture = Lecture 3: Determinant :-)